FCC-ue and Muon Collider: $HH \to 4b$ and $\nu\bar{\nu}4b$ Rates

Updated unpolarized beam configurations and preliminary LO parton-level cross sections

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Updated collider configurations. This comparison uses $\mu^+(5\,\text{TeV}) + e^-(183\,\text{GeV})$ at FCC-ue ($\sqrt{s} = 1.913\,\text{TeV}$) and $\mu^+(1\,\text{TeV}) + \mu^-(1\,\text{TeV})$ at the muon collider ($\sqrt{s} = 2.000\,\text{TeV}$). Both beams are **unpolarized** in the present results

Beam-energy configurations

Collider	**Incoming beams**	$\sqrt{s}$
FCC-ue (μ^+e^-)	$E_{\mu^+} = 5.0\,\text{TeV}$, $E_{e^-} = 183\,\text{GeV}$	1.913 TeV
Muon collider ($\mu^+\mu^-$)	$E_{\mu^+} = E_{\mu^-} = 1.0\,\text{TeV}$	2.000 TeV

Fiducial selection

- All four final-state $b/\bar{b}$ partons satisfy $|\eta_b^{\text{lab}}| < 5$.
- Form two *disjoint*, opposite-charge $b\bar{b}$ pairs. Accept the event if either distinct pairing has *both* masses in $110 \le m_{b\bar{b}}^{(1,2)} \le 130\,\text{GeV}$; count each event once.
- **No minimum p_T cut** is applied (in particular, the optional 5 GeV threshold is *not* used).
- These are parton-level estimates (no showering, jet clustering, tagging or detector simulation).

Cross sections before and after cuts

All entries are in fb. After-cuts errors are approximate Monte Carlo statistical uncertainties only.

Collider / beams	**Final state**	**Before cuts**	**After cuts $\pm\delta_{\text{MC}}$**
FCC-ue μ^+e^-	$\nu_e\bar{\nu}_\mu HH \to \nu_e\bar{\nu}_\mu\, 4b$	0.274277	**0.264782 ± 0.000430**
	$\nu_e\bar{\nu}_\mu\, 4b$	2.532894	**0.281573 ± 0.004625**
Muon collider $\mu^+\mu^-$	$\sum_{\ell=e,\mu,\tau} \nu_\ell\bar{\nu}_\ell\, HH(\to 4b)$	0.307077	**0.306240 ± 0.000542**
	$\sum_{\ell=e,\mu,\tau} \nu_\ell\bar{\nu}_\ell\, 4b$	2.745109	**0.324799 ± 0.005085**